AMANDEEP

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Senior/Lead Software Engineer — 5G Telecom Systems (C++)

PROFESSIONAL SUMMARY

Senior/Lead Software Engineer with 7+ years architecting carrier-grade 5G telecom platforms (CBRS, ODP, gNB CU/DU). Deep C++ systems programming expertise across OAM (CM/PM/SWM), FCAPS, and multi-vendor network management, with a strong record of production-readiness ownership — static/dynamic analysis, memory-leak resolution, and multi-threaded system debugging. Complements this with hands-on Kubernetes, Helm, and container orchestration experience, bridging core telecom engineering with modern cloud-native platform architecture.

CORE SKILLS

Telecom: 5G NR • OAM (CM/PM/SWM) • FCAPS • gNB CU/DU • CBRS/CBSD • ODP • NETCONF • YANG

Languages: C++ (advanced) • Python • Shell Scripting

C++ & Systems: Object-Oriented Programming (OOP) • Data Structures & Algorithms • Design Patterns • STL • Memory Management • Multithreading • Linux

Debugging & Quality: GDB • Valgrind • Klocwork • SonarQube • Memory Leak & Core Dump Analysis

Data, APIs & Monitoring: MongoDB • Prometheus • REST APIs

Cloud & Platform: Kubernetes • Docker • Helm • StatefulSets • containerd • Deployment Automation

DevOps & Tooling: Git • GitHub • GitLab • Bitbucket • Jenkins • CI/CD • Robot Framework • GoogleTest • Jira

PROFESSIONAL EXPERIENCE

Senior Engineer / Lead Engineer — Radisys Corporation, Bangalore, India *Oct 2022 – Present*

- **System Ownership:** Owned end-to-end design and implementation of production-critical 5G OAM CM/PM components across gNB CU/DU, supporting high-throughput, low-latency telecom workloads.
- **PM Architecture & Scalability:** Designed modular PM architecture using counter-type-based decomposition, enabling independent feature evolution and reducing coupling across measurement paths.
- **Performance & Efficiency:** Optimized PM execution paths and data-handling logic to reduce redundant processing and improve runtime efficiency under sustained traffic conditions.
- **DSON Reverse-Path Integration:** Implemented DSON-initiated reverse-path update support across NR (5G) and EUTRA (LTE) networks, ensuring seamless inter-RAT interoperability and configuration consistency.
- **Code Quality & Reliability:** Improved production stability by systematically resolving Klocwork, Valgrind, and SonarQube findings, including memory leaks, race conditions, and null dereferences.
- **Production Readiness:** Maintained production-grade C++ standards through deep code reviews, static/dynamic analysis enforcement, and proactive defect prevention before release cutoffs.
- **Cloud-Native Platform Architecture:** Led modernization of the Open Domain Proxy (ODP) platform to a fully containerized Kubernetes architecture — StatefulSets, persistent volumes, service discovery, and production-ready automation.
- **High Availability (MI-HA):** Contributed to Multi-Instance High Availability design for ODP services, including failover handling and MongoDB replica-set integration.

Software Engineer / Sr. Software Engineer — Capgemini Engineering, Gurgaon, India *Jul 2019 – Sep 2022*

- **Protocol Engineering:** Developed a bidirectional SIP ↔ INAP ↔ CAMEL protocol converter service, translating requests from the network and responses back to it across heterogeneous telecom systems.
- **CDR & Call-Flow Tooling:** Built CDR (Call Data Record) processing and GUI modules with unit testing; performed trace and flow analysis to link and validate call flows end-to-end across services.
- **AAA & Subscriber Systems:** Built a new RADIUS-based AAA server for subscriber authentication and accounting of traffic originated in TIM's fixed network, across server and site; developed a companion authorization service for data sessions initiated by Nokia GGSN and CMG, implemented on top of the RTCC subscriber database using RADIUS authorization.

- **Nokia NetAct / NetApps:** Contributed to NetAct, a multi-vendor, multi-technology network management system spanning NMS and EMS functions, and NetApps, an Intelligent Network platform powering subscriber services such as Freephone, Televoting, and Personal Number.
- **Quality, Audits & Automation:** Participated in Quality, Configuration Management, and Technical Audits; built a Shell-scripting-based automation tool to capture alarms and traces across multiple services.
- **Automation, Security & Release Delivery:** Automated Critical Patch Updates across clustered Oracle database environments using Robot Framework; built Python tooling for rapid Log4j vulnerability detection and mitigation; delivered multiple Priority Packages (PP) and Corrective Deployment Actions (CDA) for customer production releases.

EDUCATION

B. Tech, Electronics & Electrical Engineering — KIIT University, Bhubaneswar, India | 2015 – 2019

CERTIFICATIONS

- AWS Cloud Technical Essentials — Coursera
- Introduction to Agile Development and Scrum — Coursera